

WAZUH SECURITY MONITORING PROJECT

Windows 11 Endpoint Detection and SOC Investigation

PROJECT DOCUMENTATION

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Security Monitoring Platform: Wazuh

Endpoint: Microsoft Windows 11

Virtualization Platform: UTM

Wazuh Agent: king_windows (Agent ID 002)

Manager: kingsolomon

1. Executive Summary

This project demonstrates the deployment and practical use of Wazuh as a security monitoring and detection platform for a Microsoft Windows 11 endpoint running as a virtual machine in UTM. The project focused on validating endpoint telemetry, security-event collection, Wazuh alert generation, and SOC-style investigation.

Eight controlled security-monitoring scenarios were performed. These included failed authentication, user account creation and deletion, process creation, PowerShell process discovery, account lockout, privileged logon activity, and Windows audit policy modification. The tests generated Windows Security or PowerShell events that were collected by the Wazuh agent and analyzed by the Wazuh manager and dashboard.

The results demonstrate that the environment can detect a range of authentication, account-management, process, PowerShell, privilege, and audit-policy activities and associate several events with MITRE ATT&CK techniques.

2. Project Objectives

- Validate communication between the Windows 11 endpoint and Wazuh.
- Collect Windows security and PowerShell events using the Wazuh agent.
- Generate controlled security events on the endpoint.
- Verify that Wazuh detects and classifies the events.
- Investigate Wazuh alert severity, rules, event IDs, and relevant fields.
- Map detected activity to MITRE ATT&CK where provided by Wazuh.
- Demonstrate a practical SOC workflow from event generation through investigation.

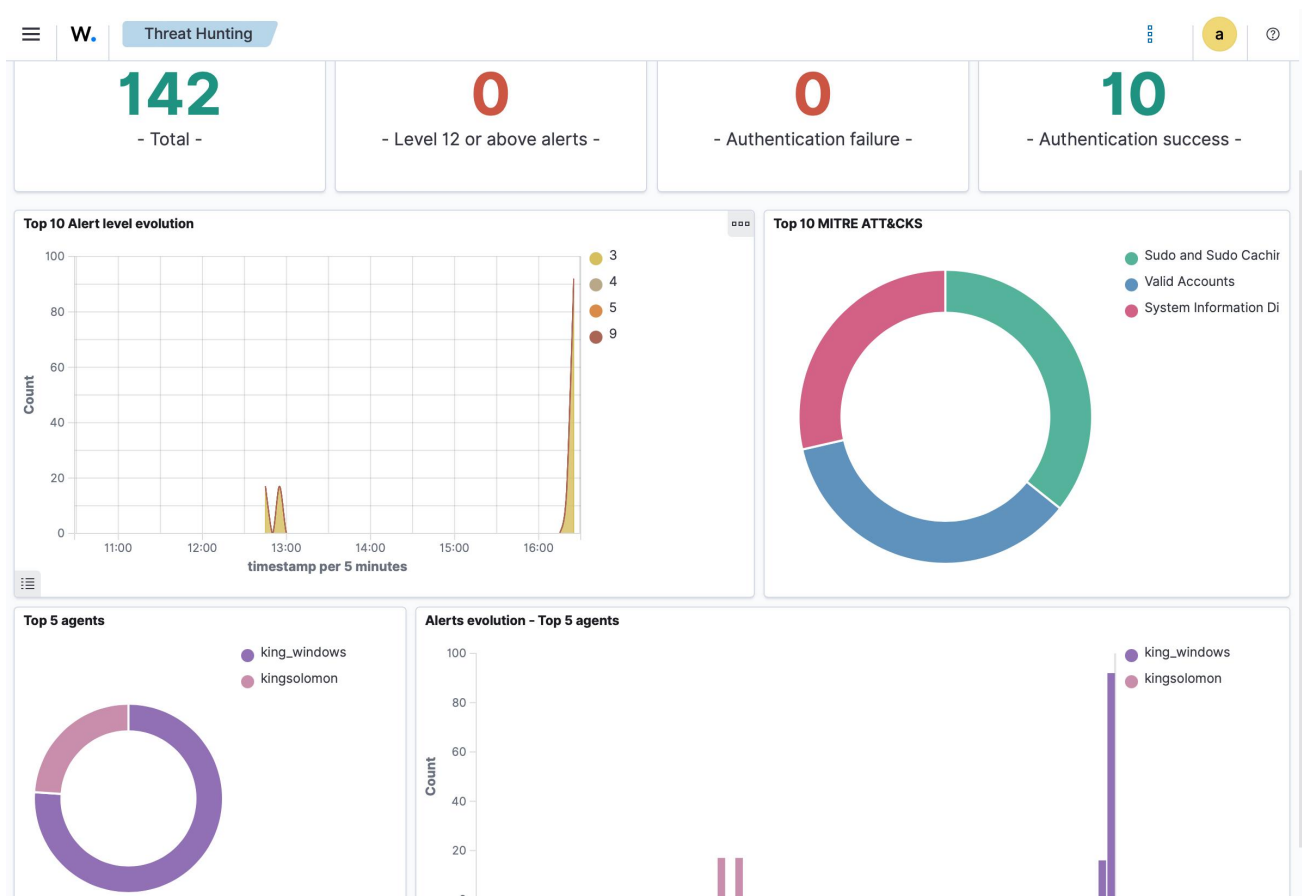
3. Lab Environment

Component	Details
Host/Virtualization	MacBook with UTM
Target Endpoint	Microsoft Windows 11
Wazuh Agent	king_windows
Agent ID	002
Agent IP	192.168.64.6
Wazuh Manager	kingsolomon
Wazuh Alert Index	wazuh-alerts-4.x-*

4. Methodology

Each test followed the same basic SOC validation process: (1) generate a controlled activity on the Windows endpoint, (2) allow the Wazuh agent to collect the relevant event, (3) locate the resulting alert in the Wazuh dashboard, (4) inspect the event ID, rule, severity and supporting fields, and (5) interpret the activity in context.

Because the activities were intentionally generated by the project operator, the alerts are treated as validation events rather than evidence of an actual compromise.



5. Detection Test Summary

Test	Windows Event	Activity	Wazuh Rule	Level	Key Classification
1	4625	Failed Logon	60122	5	Authentication failure
2	4720	User Account Created	60109	8	Account manipulation / Persistence
3	4726	User Account Deleted	60111	8	Account manipulation / Impact
4	4688	Process Creation	67027	3	Process monitoring
5	4104	PowerShell Script Block / Process Discovery	91815	4	T1057 / Discovery
6	4740	Account Lockout	60115	9	T1110 / Brute Force; T1531
7	4672	Special Privileges Assigned	67028	3	Privileged logon monitoring
8	4719	Audit Policy Change	60112	8	Audit-policy monitoring

6. Detailed Detection Tests

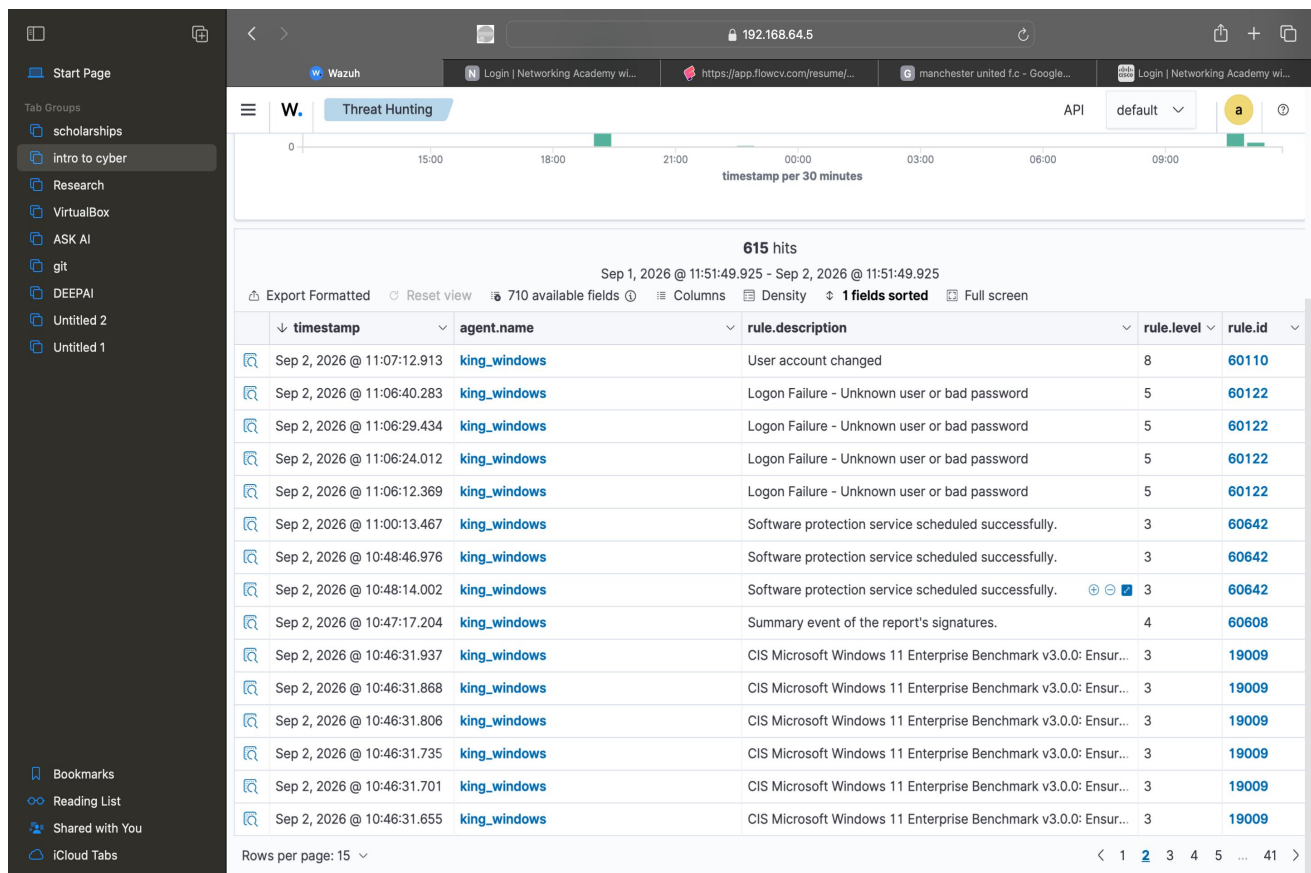
Test 1 — Event ID 4625: Failed Logon

A Windows 11 endpoint generated four Event ID 4625 authentication failures. Wazuh rule 60122 detected the activity at severity level 5. The logon type was interactive (Type 2), and the source address was 127.0.0.1, indicating that the authentication attempts originated locally rather than from a remote network host. The activity was intentionally generated as a controlled test to validate Wazuh's authentication monitoring capability.

data.win.eventdata.subjectUserSid	S-1-5-18
data.win.eventdata.targetUserSid	S-1-0-0
data.win.system.channel	Security
data.win.system.computer	king_windows
data.win.system.eventID	4625
data.win.system.eventRecordID	18678
data.win.system.keywords	0x8010000000000000
data.win.system.level	0
data.win.system.message	"An account failed to log on." Subject: Security ID: S-1-5-18 Account Name: KING_WINDOWS
data.win.system.opcode	0
data.win.system.processID	824
data.win.system.providerGuid	{54849625-5478-4994-a5ba-3e3b0328c30d}
data.win.system.providerName	Microsoft-Windows-Security-Auditing
data.win.system.severityValue	AUDIT_FAILURE
data.win.system.systemTime	2026-09-02T11:06:38.0248116Z
data.win.system.task	12544
data.win.system.threadID	1368

Detection chain: Controlled failed login → Windows Event 4625 → Wazuh Agent → Wazuh Manager → Wazuh Alert → SOC investigation

SOC interpretation: The four failed attempts were detected successfully. The localhost source is important context because it indicates the test was local rather than a remote network attack.



Test 2 — Event ID 4720: User Account Created

A Windows 11 endpoint generated a successful user account creation event, identified as Windows Event ID 4720. Wazuh rule 60109 detected the activity at severity level 8. The account wazuh_test was created by the user sulay on the local system. Wazuh mapped the activity to MITRE ATT&CK T1098 (Account Manipulation) under the Persistence tactic. The activity was intentionally generated as a controlled test to validate Wazuh's ability to detect and monitor user account creation events.

SOC interpretation: This test demonstrates why account creation is security-relevant: unauthorized accounts can be used to maintain access to a system. In this lab, the account was intentionally created.

Test 3 — Event ID 4726: User Account Deleted

A Windows 11 endpoint generated a successful user account deletion event, identified as Windows Event ID 4726. Wazuh rule 60111 detected the activity at severity level 8. The account wazuh_test was deleted by the user sulay on the local system. Wazuh mapped the activity to MITRE ATT&CK T1098 (Account Manipulation) and T1531 (Account Access Removal), under the Persistence and Impact tactics. The activity was intentionally generated as a controlled test to validate Wazuh's ability to detect and monitor user account deletion events.

SOC interpretation: The deletion event provides an auditable record of account removal and complements the account-creation test.

Test 4 — Event ID 4688: Process Creation

A Windows 11 endpoint generated a successful process creation event, identified as Windows Event ID 4688. Wazuh rule 67027 detected the activity at severity level 3. The process notepad.exe was created by cmd.exe under the user account sulay. The event was recorded as an audit success and the process creation rule fired three times. The activity was intentionally generated as a controlled test to validate Wazuh's ability to monitor and detect newly created processes.

SOC interpretation: Process creation telemetry is useful for identifying suspicious or unexpected programs and parent-child process relationships.

Test 5 — Event ID 4104: PowerShell Process Discovery

A Windows 11 endpoint generated a PowerShell Script Block Logging event, identified as Windows Event ID 4104. Wazuh rule 91815 detected the activity at severity level 4 with the description 'Powershell executing process discovery.' The script executed was Get-Process | Select-Object -First 3, which queries running processes on the endpoint. Wazuh mapped the activity to MITRE ATT&CK T1057 (Process Discovery) under the Discovery tactic. The event was recorded from the Microsoft-Windows-PowerShell/Operational channel and was intentionally generated as a controlled test to validate Wazuh-based detection of PowerShell process discovery activity.

SOC interpretation: This is a strong SOC test because it demonstrates both PowerShell Script Block Logging and Wazuh's ATT&CK classification of process discovery.

Test 6 — Event ID 4740: Account Lockout

A Windows 11 endpoint generated a successful account lockout event, identified as Windows Event ID 4740. Wazuh rule 60115 detected the activity at severity level 9 with the description 'User account locked out (multiple login errors).' The account wazuh_lockout was locked out on the local endpoint king_windows after multiple unsuccessful authentication attempts. Wazuh mapped the activity to MITRE ATT&CK T1110 (Brute Force) under the Credential Access tactic and T1531 (Account Access Removal) under the Impact tactic. The activity was intentionally generated as a controlled test to validate Wazuh's ability to detect and monitor account lockout activity resulting from repeated authentication failures.

SOC interpretation: This was the highest-severity test in the project at Level 9 and demonstrates detection of repeated authentication failures leading to account lockout.

Test 7 — Event ID 4672: Special Privileges Assigned

A Windows 11 endpoint generated a successful privileged logon event, identified as Windows Event ID 4672. Wazuh rule 67028 detected the activity at severity level 3 with the description 'Special privileges assigned to new logon.' The event was associated with the account sulaymansanyang79@gmail.com and included elevated privileges including SeDebugPrivilege, SeBackupPrivilege, SeRestorePrivilege, and SeTakeOwnershipPrivilege. Wazuh mapped the activity to MITRE ATT&CK T1484 (Domain or Tenant Policy Modification) under the Defense Evasion and Privilege Escalation tactics. The activity was intentionally generated as a controlled test to validate Wazuh's ability to monitor privileged logon activity.

SOC interpretation: Privileged-logon events are important because elevated privileges can enable significant changes to a Windows system. The alert should be interpreted with the identity and context of the administrator.

Test 8 — Event ID 4719: Audit Policy Change

A Windows 11 endpoint generated Windows Event ID 4719 after the Process Creation audit policy was modified. Wazuh rule 60112 detected the activity at severity level 8 with the description 'Windows Audit Policy changed.' The event was associated with the sulay account on king_windows and recorded a change to the Detailed Tracking → Process Creation audit subcategory, with the change indicating that Success auditing was added. The activity was intentionally generated as a controlled test to validate Wazuh's ability to monitor Windows audit policy changes.

SOC interpretation: Audit-policy changes are security-relevant because attackers may attempt to weaken or alter logging. In this controlled test, the change was intentional.

7. SOC Analysis and Findings

- Wazuh successfully received and analyzed Windows authentication, account-management, process, PowerShell, privilege, and audit-policy events.
- The project demonstrated different alert severities, ranging from Level 3 to Level 9.
- Account lockout (Event 4740) produced the highest observed Wazuh severity in these tests: Level 9.
- Account creation, account deletion, and audit-policy modification generated Level 8 alerts, highlighting the importance of administrative changes.
- The failed-logon test demonstrated the importance of contextual investigation: the source address was 127.0.0.1 because the activity was generated locally.
- PowerShell Event 4104 demonstrated that Wazuh can detect script-block activity and classify a specific behavior as MITRE ATT&CK Process Discovery (T1057).
- Process creation monitoring provides visibility into executable activity and parent-child process relationships.
- Controlled tests provide repeatable evidence that the monitoring pipeline is functioning.

8. MITRE ATT&CK Mapping Observed

Technique	Name	Observed Context	Tactic
T1098	Account Manipulation	User account creation/deletion	Persistence
T1531	Account Access Removal	Account deletion / lockout classification	Impact
T1057	Process Discovery	PowerShell Get-Process activity	Discovery
T1110	Brute Force	Repeated failed authentication leading to lockout	Credential Access
T1484	Domain or Tenant Policy Modification	Privileged-logon alert classification	Defense Evasion / Privilege Escalation

9. Evidence and Alert Records

The Wazuh dashboard was used to inspect individual alert records. The following URLs were supplied as project evidence. They point to the Wazuh instance used during the lab and may require access to the same Wazuh environment.

- Test 1 — 4625: https://192.168.64.5/app/discover#/doc/wazuh-alerts-*/wazuh-alerts-4.x-2026.09.02?id=vVfMYaABbAbGevtUz0gt
- Test 2 — 4720: https://192.168.64.5/app/discover#/doc/wazuh-alerts-*/wazuh-alerts-4.x-2026.09.02?id=21foYaABbAbGevtUk0j8
- Test 3 — 4726: https://192.168.64.5/app/discover#/doc/wazuh-alerts-*/wazuh-alerts-4.x-2026.09.02?id=_FcyYqABbAbGevtUqUjf
- Test 4 — 4688: https://192.168.64.5/app/login?nextUrl=%2Fapp%2Fdiscover#/doc/wazuh-alerts-*/wazuh-alerts-4.x-2026.09.02?id=OFfxYqABbAbGevtUHEl-
- Test 5 — 4104: https://192.168.64.5/app/discover#/doc/wazuh-alerts-*/wazuh-alerts-4.x-2026.09.03?id=gtJmaaABINz09I-TLIHo
- Test 6 — 4740: https://192.168.64.5/app/discover#/doc/wazuh-alerts-*/wazuh-alerts-4.x-2026.09.03?id=sdJ0aaABINz09I-TIIHA
- Test 7 — 4672: https://192.168.64.5/app/discover#/doc/wazuh-alerts-*/wazuh-alerts-4.x-2026.09.03?id=ZtLYZ6ABINz09I-Ta05Y
- Test 8 — 4719: https://192.168.64.5/app/discover#/doc/wazuh-alerts-*/wazuh-alerts-4.x-2026.09.06?id=5tLxeKABINz09I-TzFke

10. Recommendations

- Continue monitoring Event IDs 4625, 4740, 4720, 4726, 4688, 4104, 4672, and 4719.
- Investigate repeated failed logons and account lockouts for signs of password guessing or brute-force behavior.
- Review newly created accounts and confirm that they have an authorized business purpose.
- Monitor privileged-logon activity and validate that elevated privileges are expected.
- Protect audit-policy configuration because unexpected changes can reduce visibility into later activity.
- Use Wazuh dashboards and alert filtering to correlate multiple events rather than analyzing isolated alerts only.
- For future work, add network-oriented tests and more PowerShell behaviors to broaden endpoint detection coverage.

11. Conclusion

The Wazuh project successfully demonstrated endpoint security monitoring on a Windows 11 virtual machine. Eight controlled scenarios produced observable Windows events and Wazuh alerts, confirming that the agent-to-manager-to-dashboard monitoring pipeline was operational.

The project also demonstrated practical SOC analysis: an alert must be interpreted using the event ID, account, process, source address, severity, and surrounding context. The controlled nature of the tests made it possible to validate detection capabilities while clearly distinguishing test activity from genuine malicious behavior.